

Explainable Multi-Algorithm Plagiarism & Document Similarity Analyzer

Data Structures & Algorithms — Project Presentation - Review 3

Team Members

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Project Scope

A multi-algorithm document similarity analysis framework combining exact pattern detection, fuzzy matching, fingerprinting, and explainable reporting — designed for academic and document review contexts.

NOVELTY OF DEVELOPED METHODOLOGY

Plagiarism can take many forms, making it difficult to detect with a single approach.

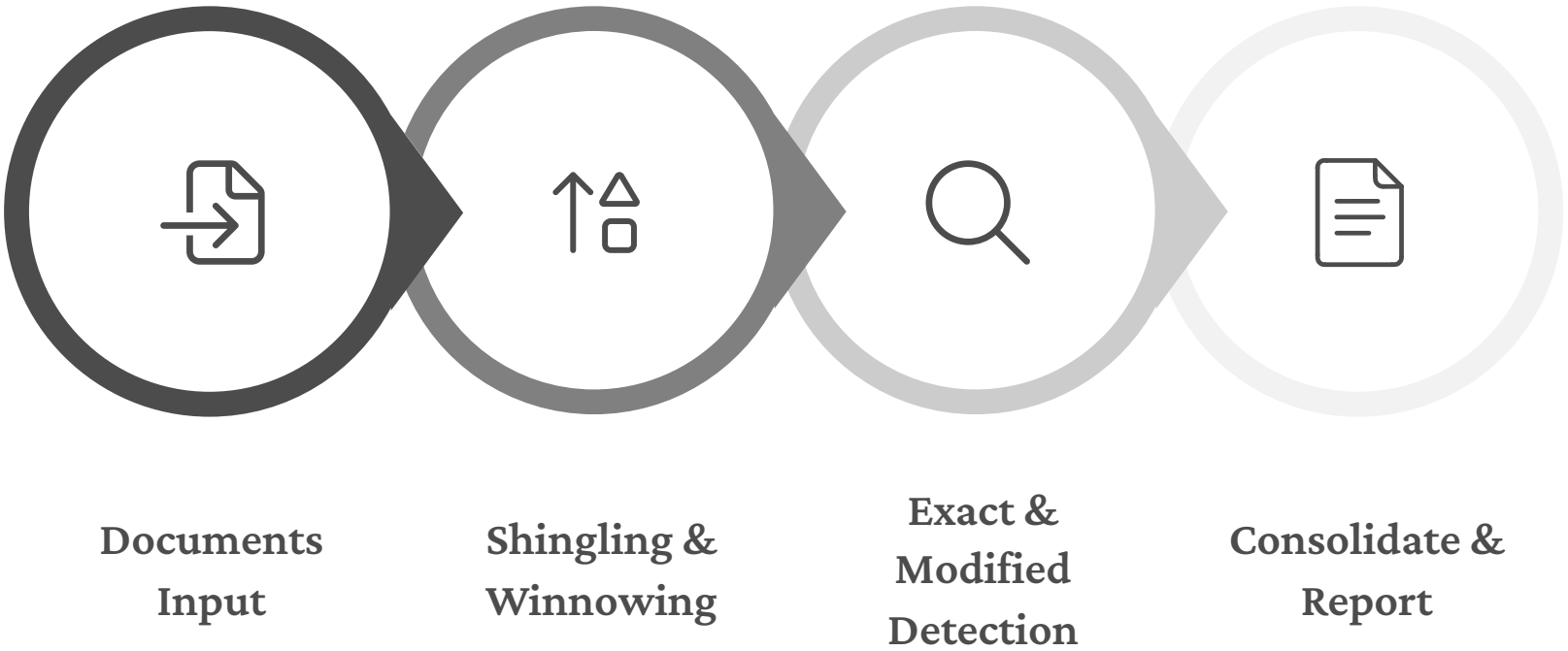
Exact Copying
Verbatim text, phrases, and sentences copied directly.

Modified Text
Paraphrased content, rearranged text, and character or word changes.

Repeated Sections
Long common passages and extended similarity across documents.

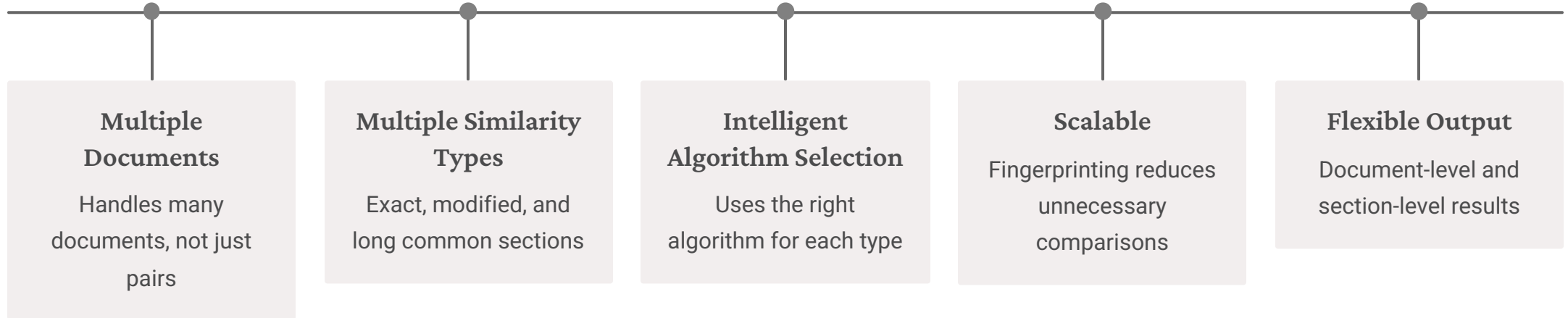
These variations make the problem difficult to solve with a single method. Our objective is to build a system that applies different algorithms to different similarity situations and provides evidence explaining what was detected, where it appeared, and why it was flagged.

The Technical Pipeline



The technical pipeline is structured to move from raw documents to explainable results in a clear sequence. Each stage narrows the comparison, detects different similarity patterns, and consolidates findings into a final report that explains the matched content and the reason for flagging it.

METHODOLOGY EVALUATION



The system adapts to different scenarios, making it flexible and efficient across various document collections and similarity types.

DISSEMINATION AND ACCESSIBILITY

Detection Accuracy

Verify correct identification of copied, modified, and common content

Efficiency

Measure execution time as document size and corpus grow

Explainability

Ensure clear matching evidence and algorithm attribution

Scalability

Stress-test with increasing numbers and sizes of documents

These four dimensions ensure our system is accurate, fast, transparent, and ready for real-world use.

APPLICATIONS AND ACCESSIBILITY

Educational Institutions

Compare assignments and reports
Verify originality before submission

Research & Academic Work

Preliminary similarity checking
Originality verification for drafts and papers

Professional Document Review

Identify overlapping content
Support compliance workflows

The system provides a simple interface for uploading multiple documents and receiving detailed similarity reports with explainable evidence.

EXPECTED SYSTEM OUTCOME

Illustrative Example of Expected System Output

Our system produces detailed, explainable similarity reports instead of just a percentage score.

Document Similarity Matrix

Document Comparison Matrix			
=====			
DOCUMENT SIMILARITY MATRIX			
=====			
	Document 1	Document 2	Document 3
Document 1	100.00%	100.00%	57.62%
Document 2	100.00%	100.00%	57.62%
Document 3	57.62%	57.62%	100.00%
=====			

Report Components

	Exact Matches KMP / Rabin-Karp
	Modified Matches Edit Distance + DP
	Longest Common Sections Suffix Array + LCP
	Matching Evidence Highlighted Text & Context

Sample values for demonstration only. Final similarity scores will be generated after implementation and testing.

Thank You

Explainable Multi-Algorithm Plagiarism & Document Similarity Analyzer



Exact Detection

KMP · Z-Algorithm · Rabin-Karp



Fuzzy Matching

Edit Distance + DP



Common Content

Suffix Array + LCP



Fingerprinting

Shingling + Winnowing

Team

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Section & Group

Section A7 · Group G-9 · Data Structures & Algorithms